Fit and radio: parts need space beyond their outlines

The reading that goes with the cardboard mock-up. “The rectangles fit” is only the start: real parts vary in size, plugs need a path in and out, and the antenna reacts to everything near it, even things it never touches.

Where this sits

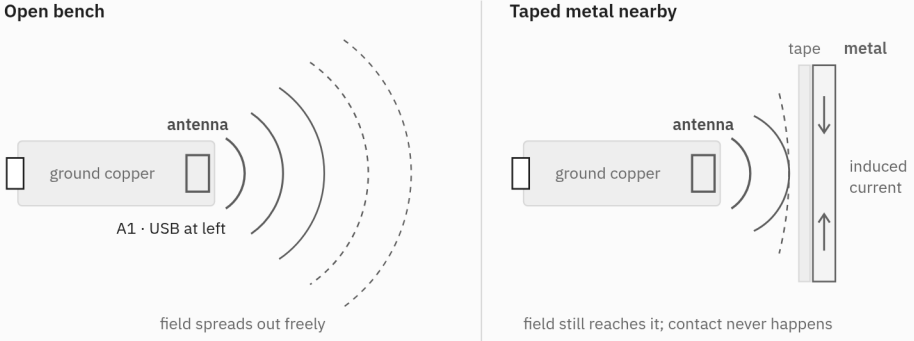
1–2 · done	3 · done	4a · now	4b–4e · next
Prototype + cloud	**It speaks**	**Mock-up + this reading**	**Case**
USB prototype, Wi-Fi, assistant replies	amplifier, cap 45	why the fit rules exist	paint test, frame design, brass, wiring

1 The antenna includes everything around it

An antenna turns fast-changing current into radio waves, and incoming waves back into current. It is not just the small ceramic chip at the end of A1. The radio, the board's ground copper, the USB cable, your hand and any nearby metal form **one system**.

The antenna's field pushes currents around in nearby metal **without any contact**. Those induced currents act back on the antenna:

DETUNE	REFLECT	ABSORB	RE-RADIATE
the antenna works best at the wrong frequency	signal is shadowed in some directions	energy is lost as heat in the metal	the metal becomes an unplanned second antenna

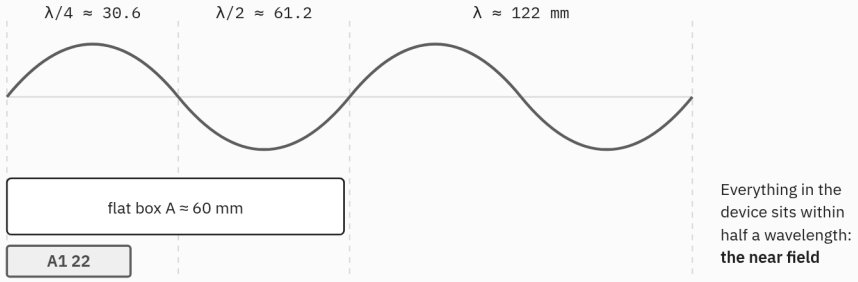


Tape stops a short circuit, not a field. Insulated metal is just as visible to the radio as bare metal. Conceptual drawing, not a keepout map or a real field plot.

> **YOUR BOARD ALREADY PROVED THIS**
>
> A1's chip antenna sits too close to its own ground copper, so much of the transmitter's output reflects back instead of radiating. Its setup hotspot was invisible at the default 20 dBm and appeared only once the transmit power was capped at 8.5 dBm. A1 has very little radio margin to begin with, so the enclosure must not take any more away.

2 Wavelength gives you scale, not rules

```
wavelength = speed of light / frequency
           = 3.00×10⁸ m/s / 2.45×10⁹ Hz ≈ 0.122 m = 122 mm
quarter wavelength ≈ 30.6 mm      half wavelength ≈ 61.2 mm
```



Drawn to scale (4 px = 1 mm). The whole pocket device is smaller than half a wavelength, so every part, wire and frame member is in the antenna's near field, where objects couple strongly even when they are not between the antenna and the router.

These numbers tell you how big the problem is. They do **not** mean “never use a 30 mm piece of brass.” Whether a piece of metal resonates depends on its shape, width, bends, material, how it couples to the board and what surrounds it, not on its length alone.

3 The 15 mm rule is a starting point

Espressif, who make the ESP32-C3, recommend that a module's antenna stick out past the board it is mounted on. Where it can't, they ask for **at least 15 mm of clearance** around the antenna area with no copper, traces or parts.

That is good first-layout guidance for their own antenna design. It does not certify a SuperMini clone, and it does not promise good Wi-Fi beside a brass frame. For the mock-up it turns into three habits:

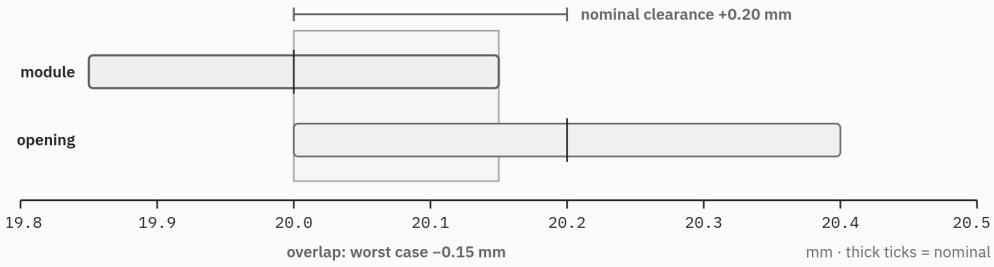
- Put **A1's antenna end against an outside edge**. This is rule 1 on the mock-up page.
- Keep the speaker wires, the amplifier and the screen's cable away from that end.
- Leave room to **try more than one antenna position** before anything becomes permanent.

4 Nominal size isn't the same as clearance

WORD	MEANING
Nominal	the design size, the number on the drawing
Tolerance	how far a real part may differ from it
Clearance	the space left over; negative means the parts collide

Example: an opening of `20.2 ± 0.2 mm` and a module of `20.0 ± 0.15 mm`.

```
nominal clearance    = 20.2 − 20.0                    = +0.20 mm   looks fine
worst-case clearance = (20.2 − 0.2) − (20.0 + 0.15)   = −0.15 mm   does not fit
```



The nominal drawing fits, the extremes do not. The shaded band is where a large module meets a small opening. Stacking parts adds more of these ranges together: board thickness, pin headers, tape, the lid. That sum is a **tolerance stack**, and it is why the stacked arrangement B carries more risk than flat arrangement A.

This is also why the mock-up includes things that are not board outlines: the **USB plug body and its bend room**, the button's travel, the mic hole and the speaker opening. Most first layouts fail on the USB plug or the antenna, not on whether the boards fit. Your caliper's digits do not tell you a part's tolerance either. Measure the real part, and measure more than once.

5 Radio is noisy, so compare one change at a time

When you later test the brass frame, a single signal-strength reading (RSSI) proves almost nothing. Reflections in a room add and cancel, so moving the board a few centimetres can change the reading by several dB. A useful comparison looks like this:

1. Fix the board on a non-metal stand. Record the router position, distance, channel, cable route and firmware.
2. Run several equal-length **baseline** tests with no metal nearby.
3. Change **one thing**, for example a brass piece at 5, 10, 15 and 30 mm, and repeat the same tests.
4. Compare the **typical value and spread**, plus dropped packets and reconnects, never just the best reading.

5 Re-run the baseline at the end. If it moved, the room changed and you need more runs.

Write the expectation down first, so the test can prove it wrong: “Moving the antenna from 5 mm to 15 mm away from the nearest brass improves typical RSSI without more dropped packets.” That is a claim you can test. “15 mm guarantees good Wi-Fi” is not.

> **WHAT THE PAPER MOCK-UP CAN AND CANNOT TELL YOU**
>
> Cardboard checks **space**: fit, plug reach, openings, wire room. It says nothing about **radio**. Only a test of the real geometry, done as above, can.

Check yourself

Tape separates A1 from a metal lid so they can't touch. Will Wi-Fi behave as it did on the open bench?

No. Insulation stops direct contact, not the field. The lid can still detune and shadow the antenna, so only a controlled test with the lid in place answers it.

Why doesn't a 30.6 mm quarter wavelength make every 30 mm brass bar an antenna?

Resonance also depends on shape, width, bends, material, end effects, coupling, losses and surroundings. Length alone does not decide it.

What does Espressif's 15 mm recommendation establish, and what doesn't it?

It is the maker's starting layout guidance for its own antenna design. It does not certify a clone board or guarantee performance beside a metal frame.

Nominal clearance is +0.2 mm. Can the parts still collide?

Yes. If the opening comes out at its smallest and the part at its largest, the worst-case clearance can be negative, as in the example above.

Name two things to record besides RSSI in a frame test.

Dropped packets or retries, and disconnects with reconnect time. Latency or throughput also work.

Read more

- **Short version:** `plan/concepts/14-fit-and-radio.md`
- **Deep version:** `edu/fundamentals/11-rf-emc-antennas-and-metal-frame.md`. Read “The manufacturer's placement rule”, “RF performance is statistical” and “Layout hypotheses must be testable” closely. The battery, converter and frame-grounding parts apply to a portable version, which is out of scope for this USB prototype, so skim them.
- **Next:** the tolerance section of `edu/fundamentals/12-soldering-mechanics-insulation-tolerance.md`, the deeper version of section 4.

Sources. Built from `plan/concepts/14-fit-and-radio.md` and `edu/fundamentals/11-rf-emc-antennas-and-metal-frame.md`. A1's 8.5 dBm transmit-power cap comes from `plan/FAST_TRACK_THEORY.md` Step 10. The 15 mm guidance is Espressif's ESP32-C3 PCB layout guidance. Box and board sizes come from the cardboard mock-up page.

What is and is not verified. This is theory. No radio test of any enclosure has been run. The field drawing is conceptual; the wavelength figure is to scale; the tolerance example uses illustrative numbers, not your parts' real tolerances. If this page and the repository differ, the repository wins.